

Kleene: Relational Algebra for Recursive Language-Model Calls

A planner, a budget and a replay-gated playbook for agents that write SQL

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Abstract

Recursive Language Models (RLMs) let a model treat its context as a variable in a REPL and call itself on slices of it. Every published harness gives the model Python for that REPL. We argue that for this job SQL is the better language, and we present KLEENE, a Rust harness in which the model’s only action is to write CallSQL: a PostgreSQL-flavoured dialect in which model calls are scalar and table functions, tools are table functions or **CALL** statements, delegation is a lateral table call that opens a child session, search is a recursive CTE with a beam limit, and the answer is a relation. Because the work is a query, three things follow that a tool-calling loop cannot offer. A *planner* sees the cost of every call predicate and applies eight rewrite rules (cheap-first, memo dedupe, batching, proxy cascades, semi-joins for **EXISTS**, beam-limited recursion, join ordering with branch-and-bound, volatility fences) with a cost model that **EXPLAIN** shows before any money is spent. *Budgets* in calls, tokens, dollars, depth and wall clock are query semantics: a statement whose estimate exceeds what remains is refused with its plan. *Learning* is rows in DuckDB: solved SQL becomes a playbook only after a replay evaluation with the memo bypassed wins on solves and cost, and every adoption is tied to the evaluation that admitted it. We describe the algebra, the system and a benchmark harness with seven task packs run under three modes (learning, frozen, and a plain tool-calling agent on the same provider and budgets). On the first four packs Claude Opus 5.5 solves every task in every mode, so they separate on cost only: the playbook cuts calls per task by 16–39%, the replay gate costs more than the playbook saves at twenty tasks a pack, and the one-call baseline is cheapest wherever the context fits in a prompt. On three harder packs, Claude Haiku 4.5 separates on accuracy: on rubric-graded memos learning beats frozen 11/20 to 8/20 at a fifth of the calls, while on code-editing episodes both KLEENE modes solve 2/12 and the plain agent none. We report these numbers with their confounds (shared memo between modes, a stopped sweep, one run per cell) and the fixes they forced.

1 Introduction

Recursive Language Models [42] change what an agent harness hands the model. Instead of a prompt that holds the whole context, the root model sees only the task; the context lives as a variable in a REPL, and the model writes code that peeks at it, partitions it, and calls `llm_query` or `rlm_query` on slices. The strategies that emerge (peek, grep, partition-and-map, summarise, solve in code) are what one would write by hand, and the paper reports large gains on long-context aggregation at cost parity. The official harness and every reimplementations we know of give the model Python for that REPL. A reproduction study [39] found that recursion beyond depth one already degrades on current models, through format collapse and runaway sub-calls: the freedom of a general-purpose language is also the freedom to spend without a plan.

KLEENE starts from a different premise. The REPL is a relational engine and the model writes SQL. Three properties of SQL make it a better fit for exactly this job.

1. **It is declarative.** The model says *what* relation it wants. The harness decides how many calls to make, in what order, concurrently or not, and on which model. That is precisely the freedom a planner needs and that an imperative loop denies it.
2. **It has an algebra with known complexity.** Select-project-join is NP-complete in combined complexity [6], full relational algebra is PSPACE-complete [38], and `WITH RECURSIVE` reaches Datalog and beyond [8], up to Turing completeness with `UNION ALL` [9]. Every statement the model writes can be labelled with the fragment it lives in, and join ordering over call predicates is a combinatorial search the planner deliberately confronts [14].
3. **The trace is a relation.** Calls, tokens, dollars and depths land in DuckDB tables [30]. The model, the harness and the person inspect execution with the language they drive it with.

Once the work is a query, a planner, budgets and learning fall out as database problems rather than prompt engineering. This report describes the resulting system and what it has measured so far. Our contributions are:

- CallSQL, a dialect in which model calls, tools, delegation and budgets are first-class (§2), and a *call algebra* that attaches a call kind and a cost signature to any relational operator that evaluates a call (§3).
- A planner with eight rewrite rules and a cost model that mirrors the executor’s short-circuit semantics, exposed through `EXPLAIN` and used to refuse statements a budget cannot pay for (§3.1).
- A session loop with memoised calls, concurrent child sessions and a persistent store in which every learned object (playbook SQL, ratings, generator dials, refined prompt functions) is a versioned table adopted only after a replay evaluation (§4, §5).
- A benchmark harness with seven task packs, three modes that isolate the playbook and the SQL abstraction respectively, code and rubric oracles, and record-and-replay (§6); and the results of two real-model runs, reported with their confounds (§7).

What this report does *not* claim is a benchmark win. The first four packs were sized for a weaker model and saturate under Opus 5.5; the harder packs separate on accuracy but have been run once, partially, on Haiku 4.5. The numbers say where the abstraction pays for itself and where it does not, and §9 lists what remains before a stronger claim can be made.

2 CallSQL: the language the model writes

Everything the model does in a session is one or more CallSQL statements in a single `sql` fence. The relational core is a PostgreSQL-flavoured subset: `SELECT` with CTEs and `WITH RECURSIVE`, inner and lateral joins, `EXISTS`, `IN`, scalar subqueries, grouping, ordering, set operations, `VALUES`, `CREATE TABLE ... AS`, `INSERT` and `DROP`, over the types `BOOLEAN`, `BIGINT`, `DOUBLE`, `TEXT`, `JSON` and `VECTOR`. Three-valued logic short-circuits `AND` and `OR` left to right, which the planner relies on. The core is tested differentially against DuckDB, recursive CTEs included (§7.1). On top of it sit five additions.

Context as a table. A run that is given a context starts with `ctx(ordinal, text)`, one row per paragraph. The root model sees the task, the catalog and the row count, not the text. It peeks with `SELECT ... LIMIT`, filters with `grep` or `LIKE`, partitions with `chunks`, and never has to hold the context in its prompt.

Model calls as functions. The scalar builtins `llm(prompt)` (with optional alias and effort arguments), `llm_bool(prompt)` and `llm_json(prompt, schema)` each cost one call per distinct argument tuple. `expand(text, n)` is a table builtin that returns n successors. Prompt-defined functions give a call a name, a return type, a model tier and a volatility:

```
CREATE FUNCTION mentions_hours(note TEXT) RETURNS BOOLEAN
  AS PROMPT 'Does this note record hours worked? {note}'
  MODEL 'proxy' BATCH 20 IMMUTABLE;

CREATE FUNCTION is_valid(c TEXT) RETURNS BOOLEAN
  AS PROMPT 'Is {c} a valid answer to the task?'
  MODEL 'worker' PROXY plausibility THRESHOLDS (0.2, 0.8);
```

MODEL picks a routing alias (`root`, `worker`, `proxy`, `judge`, or any alias the router knows). BATCH n marks a prompt batchable: the executor answers up to n distinct tuples per call, and the planner prices $\lceil \text{rows}/n \rceil$ calls. PROXY declares a cheap scorer the planner may cascade the predicate through. Bodies may also be SQL (`AS SQL (SELECT ...)`) or a shell command (`AS SHELL`); volatility defaults to IMMUTABLE for prompts, STABLE for SQL and VOLATILE for shell.

Tools. Pure tools are table functions usable in FROM: `files`, `lines`, `read`, `grep`, `chunks`, `env`, `git_log`, `git_diff`, `git_blame`. Side effects run as statements: `CALL shell(cmd)`, `CALL write_file(path, text)`, `CALL patch(path, old, new)`, `CALL web_search(q)`, and so on. A tool's output relation is rendered back to the model like any other result. Roles restrict which tools a session may use.

Delegation. `rlm(question [, context])` in a CROSS JOIN LATERAL opens one child session per input row at depth $d+1$ and returns (`answer`, `detail`). CREATE AGENT declares a role with a model, an effort, a tool list, a budget and a prompt; `spawn('role', task)` runs it. A child's catalog is restricted to its role's tools and its budget is the parent's remaining slice intersected with the role's budget.

```
CREATE TABLE parts AS
  SELECT (ordinal / 20) AS part, string_agg(text, '\n') AS chunk
  FROM ctx GROUP BY 1;

CREATE TABLE answers AS
  SELECT p.part, r.answer
  FROM parts p CROSS JOIN LATERAL rlm(
    'Total hours per project in these notes, as project: hours', p.chunk) r;
```

Finishing, budgets and plans. FINAL FROM (query) ends the session with that relation as the answer. SET adjusts effort, model, memo and the budget dimensions within a session. EXPLAIN prints the call plan (§3.1); EXPLAIN ANALYZE runs it and prints actuals beside estimates. The harness runs the planner on every statement before execution, and a statement whose estimate exceeds the remaining budget is returned to the model as a refusal carrying the plan rather than executed.

The opening example of the repository shows the shape the design is built for:

```
SELECT candidate
FROM possibilities
WHERE VERIFY(candidate)
  AND NOT EXISTS (
  SELECT 1 FROM counterexamples ce
  WHERE REFUTE(candidate, ce.text));
```

Operator	Meaning	Cost signature
λ_f (map-call)	scalar call f per distinct input tuple	$ \delta(R) \times \text{tokens}(f)$
κ_g (expand-call)	table call g per input tuple (LATERAL $g(\dots)$)	$ R \times \text{tokens}(g)$; output = branching factor
ρ_d (recurse)	rlm / spawn at depth d	the child’s whole plan
$\sigma_{\text{llm}}, \bowtie_{\text{llm}}$	filter or join whose predicate contains a call	inherits λ_f ; $ R \times S $ calls for a join unless rewritten

Table 1: Call kinds. δ is duplicate elimination; identical prompts cost one call.

VERIFY is one call per distinct candidate. The **NOT EXISTS** is an anti-semi-join that stops on the first refuting counterexample: the counterexample-guided loop of CEGIS and CEGAR [7, 34] written relationally. **EXPLAIN** says how many calls that is before they are made.

3 The call algebra

The ordinary operators are selection σ , projection π , join $\bowtie$, union $\cup$, difference, aggregation γ and the least fixpoint μ for recursive CTEs. **KLEENE** adds a *call kind* to any operator that evaluates a call expression (Table 1). The cost of an operator is the number of calls it implies times the per-alias token and dollar factors of the function it calls; selectivity of a boolean call predicate is a prior until the predicate has been observed, after which the sampled pass rate is used.

Every statement is also labelled with the fragment it lives in: CQ for select-project-join, FO for the full relational algebra, REC when a recursive term is present. The badge is a reminder that a recursive call plan is a program, not a lookup.

3.1 Rewrite rules and the cost model

The planner in **kleene-algebra** runs rules over the annotated plan until nothing changes. All eight are standard query-rewrite shapes applied to call costs.

1. **Cheap first.** $\sigma_{\text{pure}} \circ \sigma_{\text{llm}} \equiv \sigma_{\text{llm}} \circ \sigma_{\text{pure}}$: pure conjuncts run before call conjuncts, in filters and in join conditions. Because estimates mirror the executor’s left-to-right short-circuit, the saving is visible in the numbers rather than assumed.
2. **Dedupe then map.** $\lambda_f(R) \equiv \lambda_f(\delta(R)) \bowtie R$, backed by a persistent memo keyed on model and prompt fingerprint and shared across sessions and runs.
3. **Batch.** λ_f over n tuples of a **BATCH n** function becomes one call with n items, priced $\lceil \text{rows}/n \rceil$. An answer that does not hold exactly one element per item falls back to one call per tuple.
4. **Cascade.** $\sigma_{\text{oracle}}(R)$ with a declared proxy becomes $\text{score} \geq \tau_{\text{hi}} \vee (\text{score} \geq \tau_{\text{lo}} \wedge \text{oracle})$, applied when the priced cascade is cheaper; the shape is LOTUS’s [29], with thresholds declared rather than calibrated (§9).
5. **Semi-join.** **[NOT] EXISTS** with a call predicate becomes a semi or anti join that stops at the first (counter)example instead of materialising the cross product.
6. **Beam-limited recursion.** Inside μ , a trailing **ORDER BY ... LIMIT k** in the recursive term keeps the best k new rows of each round: breadth-first search with a model successor function becomes beam search of width k , and the planner simulates the fixpoint round by round to price it.

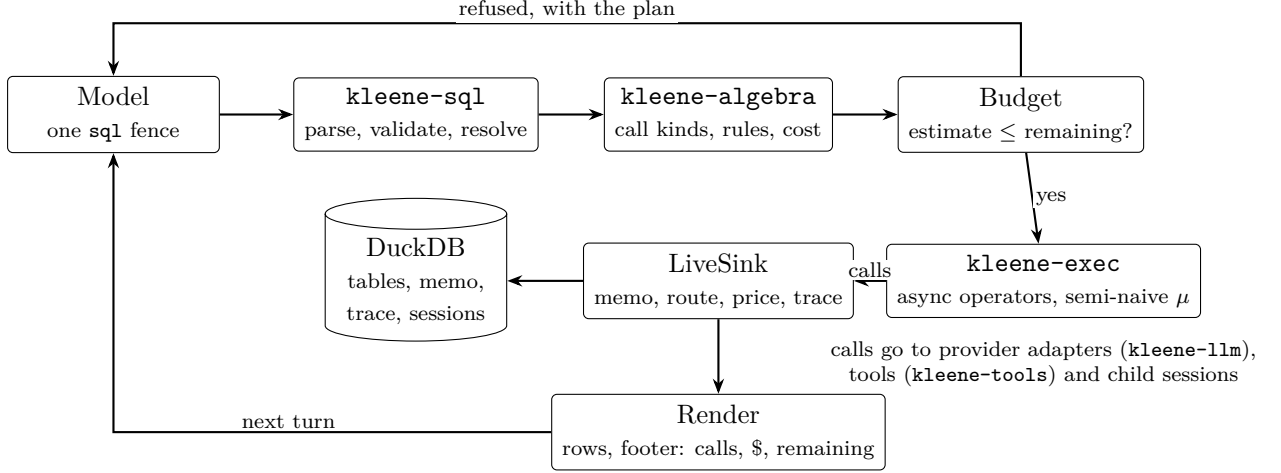


Figure 1: One turn of a session. The harness sends a cached system prefix (rules, catalog, budget) plus the transcript; the model replies with CallSQL; the statement is parsed, annotated, priced, checked against the budget, executed through a sink that memoises, routes and traces every call, and rendered back. The loop repeats until **FINAL**.

7. **Join ordering.** Dynamic programming over relation subsets with branch-and-bound, in the tradition of Selinger et al. [33] and Moerkotte and Neumann [28]: each split is priced with the predicates that first become applicable there, pure conjuncts narrowing the pairs and call conjuncts paying one call per surviving pair.
8. **Volatility fences.** Rules 1–7 move operators only across **IMMUTABLE** and **STABLE** operators. A **VOLATILE** operator (a shell command, a file write) is a fence: nothing crosses it, it is never deduplicated, and its input order is preserved. The planner trusts the label, so a side effect is never marked anything else.

Estimates use the store’s row counts, sampled selectivities, per-alias cost factors and pricing. **EXPLAIN** prints the plan as a tree with per-node rows, calls, tokens and dollars, the call kinds, the fences, the fragment badge, the rules that fired, the join-order plan space and the alternatives it priced.

Budgets as semantics. A **Budget** has calls, tokens, dollars, depth and wall clock. A child session receives the parent’s remaining slice intersected with its role’s budget, and its spending rolls up. Exceeding a budget cancels the statement, not the session, and returns a budget row to the model; a statement whose *estimate* already exceeds the remainder is refused before it runs. Depth defaults to a cap of two, following the reproduction study’s finding that deeper recursion degrades on current models [39]; it is a dial, not a constant.

4 System

KLEENE is a Rust workspace of eleven crates with a shared interface crate (**kleene-core**: values, schemas, the catalog, budgets, call kinds) that every other crate depends on. Figure 1 shows the path of one turn.

Execution. Operators are pull-based async streams of small columnar batches. Recursion runs by semi-naive evaluation [3]: each round joins only the previous round’s delta against the recursive term, so calls in that term run only on new rows, and a round is one awaitable unit the UI can show as a frontier size. Model calls are concurrent under a per-session semaphore with a provider rate limiter; ordered results are preserved.

The sink. Every scalar, table, tool or child call passes through a `LiveSink` that looks the call up in the memo, and on a miss routes it to a provider alias, prices it at the router’s list rates, charges the budget, stores the response in the memo and emits a trace event. Identical prompts are therefore free the second time, across sessions and runs; this is also how a recorded run replays offline.

Sessions and delegation. Child sessions are the same loop one level deeper with a role, a budget slice and their own table namespace; a child’s `FINAL` returns to the parent as a row. Children of one lateral call run concurrently. Session state (tables, transcript, budget) lives in the store, so killing the process and running `kleene resume` continues where it stopped; checkpointing is a property of the storage, not a feature.

Model layer. Adapters speak the Anthropic Messages and OpenAI-compatible wire formats directly over HTTPS, plus an Open Responses adapter behind a feature flag; there are no vendor SDKs. A router maps aliases to models and prices, and a `ReplayProvider` serves recorded fixtures so the test suite and the benchmark harness can run without a key. Two provider facts the benchmarks forced into the adapter are described in §7.3.

Store and trace. DuckDB holds session tables, the memo, the trace (`trace_sessions`, `trace_statements`, call events), the continual loop’s tables and the benchmark rows. `kleene trace` runs plain DuckDB SQL over all of it, and the terminal UI’s `/trace` does the same from the prompt.

Processes. `kleene run`, `repl`, `explain`, `learn` and `bench` run the harness in one process. The terminal UI talks to an engine daemon over a Unix socket with a JSONL protocol with cursors, so a reconnecting client resumes from where it left off and `kleene attach` watches the same events headless. The UI is one scrolling stream: the model’s reply streams in, each fence runs as it lands, every statement shows its rows and its cost, child sessions nest under the turn that opened them.

5 Learning as tables

The continual loop (`kleene learn`) is itself a query over a `tasks` table: pick the next pending task by curriculum policy, run a session, judge `FINAL` with the task’s oracle, record outcome and cost, repeat. Two things keep it alive: generators that keep the table non-empty, and learning that changes how the next task is attempted.

Tasks and difficulty. Generators (`sat3`, `graph`, `puzzle`, `corpus`, `repo`, `statements`, `contracts`, `logbook`, `memo`) produce tasks deterministically from a hardness dial in $[0, 1]$ and a seed, each with a code oracle; the design follows Reasoning Gym [35] and the random 3-SAT phase transition [27]. Every task and every solver configuration carries a rating moved by a Bradley–Terry model [5] over solve and fail outcomes: a task that beats strong configurations is hard, a configuration that beats hard tasks is strong. The curriculum samples the pending task nearest even odds for the current solver, steps a generator’s dial up when its solve rate exceeds 0.7 and down below 0.3, in the spirit of the learnability target of Absolute Zero Reasoner [43].

What is learned. Nothing here updates weights. Each learned object is a table with a version and a provenance pointer:

- the **playbook**: SQL that solved a task, keyed by task kind, rendered into the cached prefix as examples for similar tasks and queryable by the model;

- **refined prompt functions** (`learn refine`): new versions of `VERIFY`-style functions in a ledger with before/after snapshots and rollback;
- the **cost model**: sampled selectivities of boolean call predicates that feed the planner (today per session; §9);
- **ratings** of tasks and solver configurations, and each generator’s **dial**.

The replay gate. A candidate playbook entry is adopted only after a replay evaluation: the harness re-runs a sample of earlier tasks of the same kind in two arms, with and without the candidate, with the memo bypassed, and adopts the entry only if the arm with it wins on solves and costs no more. The sample is three tasks by default, so a gate costs six task runs per solved task. Every adoption is tied in `playbook_evals` to the evaluation that admitted it, and `learn playbook` lists reverts, so a regression can be traced to the entry that caused it. The base system prompt is immutable and learned content is appended in marked sections. §7.2 shows that at twenty tasks a pack the gate spends far more than the playbook saves; it is the price of never adopting an entry on faith.

6 Benchmark methodology

The harness’s claim is specific: a model writing CallSQL over a relational engine solves the same tasks as a tool-calling agent for fewer calls and dollars, and the harness gets better over a stream of tasks because solved SQL is kept. The benchmark harness (`kleene bench`) is built to test exactly that, so every run records three things per task: was it solved, how many model calls it took, and what it cost.

6.1 Packs

A pack is a directory with a `pack.json`: tasks with text, optional context, setup commands, a workspace source, an oracle and a difficulty prior. Table 2 lists the seven shipped packs. The synthetic packs are frozen output of the loop’s generators, so two runs see the same tasks in the same order and `bench build` regenerates them byte for byte; the two long-context packs are stored lazily as (generator, dial, seed) triples and regenerated on load. The coding pack is hand-written: four small Python projects, each a three-step *episode* in which the second and third steps run in the workspace the previous step left, and a step’s setup installs the reference solution for the previous step when its hidden tests fail, so each step is measured on its own work. The external datasets these packs imitate (OOLONG [4], Terminal-Bench [26], FinanceBench [15], CUAD [13], Harvey LAB [12], SWE-bench [16]) are not redistributed; a Harvey LAB checkout imports through `bench import-lab` with LAB’s own evaluator as the scorer of record.

6.2 Modes

Every pack runs under one of three modes, and two comparisons fall out of them.

learning KLEENE: the model writes CallSQL, the playbook is shown, candidates are recorded and adopted through the gate, ratings move.

frozen KLEENE with the same model, tools and budgets, but no playbook shown, no candidate recorded, no rating moved. The control for learning.

plain a tool-calling agent on the same provider, tools and budgets, in the ReAct shape [41]: each turn the model returns one JSON action and tool output comes back as text. The baseline for cost parity, as in the RLM paper’s framing: same model, same spend, only the abstraction differs.

Pack	n	What the model gets and must produce	Oracle	Stands in for
terminal	6	a workspace and a shell instruction; files left behind	shell	Terminal-Bench
oolong-like	20	dated meeting notes as ctx ; hours per project	exact	OOLONG aggregation
finance-synthetic	20	a synthetic income statement; one number	number	FinanceBench
legal-synthetic	20	a synthetic contract; the clause categories present	exact	CUAD / LAB
logbook-hard	30	a 30 000-token work log with corrections and distractors; one of five aggregate questions	exact	OOLONG at full size
memo-rubric	20	a 40-section agreement with an amendment and a rejected proposal; a short memo	judge	LAB drafting
coding	12	a Python project with visible tests; implement, extend, fix a bug	shell	SWE-bench episodes

Table 2: The seven shipped packs. **exact** compares row multisets, **number** the first cell within a tolerance, **shell** runs a command (here the grader’s hidden tests) and **judge** asks a separate model for **PASS** against a rubric that names every required fact and forbids the superseded values.

learning against **frozen** isolates the playbook; **frozen** against **plain** isolates the SQL abstraction. Two caveats matter for reading the results. Both KLEENE modes read the same store, so a frozen run on a store that has seen a learning run (or the reverse) is served from the memo for every prompt already sent; a clean comparison needs a fresh store per mode, which the Haiku run did not yet do (§7.3). And the plain agent parses JSON actions from text rather than using a vendor’s native tool-calling API, which keeps it provider-agnostic but is not identical to a vendor agent loop.

6.3 What is recorded

bench run writes one row per task to **evals**: run, pack, mode, position in the stream, task, solved, the oracle’s reason, calls, tokens, dollars, deepest child, turns and wall clock. **bench report** aggregates by pack and mode; **bench csv** exports the rows; **bench curve** prints the rolling solve rate of one run; **bench plot** draws five standalone SVGs (learning curve, accuracy at cost parity, calls against difficulty, planner estimate against actual, plan-space size against query shape) without a plotting library. The replay evaluations a learning run pays for are recorded in **playbook_evals** with their own dollars and are *not* in **evals**; the true cost of a learning run is the sum. **bench run -record** saves every model reply as a fixture and **-replay** serves it back, so a run reproduces offline; **-resume** continues a run that a provider error stopped.

7 Results

7.1 Deterministic measurements

Table 3 is the part of the evidence that needs no model: the test suite runs it in CI with scripted providers, code oracles and replay fixtures.

Property	Test	Result
Relational core equals DuckDB	property-based differential suite	passes, recursive CTEs included
Join ordering, two call predicates over three relations	estimate and actual calls against a counting sink	as written 12 000 estimated calls, chosen order 60; actual $\geq 10\times$ fewer with identical rows
Plan-space growth	seven-way join	665 280 bushy orders priced through 1 623 subset splits (309 pruned)
Cascade	proxy at $0.1\times$ cost, thresholds (0.2, 0.8)	100 proxy + 60 oracle calls beat 100 oracle calls by estimate; on data the oracle is asked only for the band (5 of 20)
Beam recursion	expand $\times 4$, three rounds	20 calls with beam 5 against > 100 unbounded
Continual loop	scripted solver over generated puzzles	tasks solved, playbook v1 adopted after a replay eval, ratings and dial move, restart resumes
Terminal-Bench-style pack	shell tasks from SQL	tasks solved, oracles run in fresh workspaces
Plain-agent baseline	same provider, JSON actions	tool call then final; the control for cost parity

Table 3: What the deterministic suite establishes.

7.2 Claude Opus 5.5 on the first four packs

The four original packs ran once each under the three modes on 27 September 2026 with the default routing: **root** on **claude-opus-5-5** at high effort, **worker** and **judge** on **claude-sonnet-5**, **proxy** on **claude-haiku-4-5**, priced at the router’s list rates. Table 4 gives the per-pack rows; Figure 2 the plots **bench plot** drew from the same store. **dollars** is the pack total, calls and tokens are per task, and the learning column excludes the gate.

Accuracy is saturated. Opus 5.5 solves every task of every pack in every mode. The packs were sized for a weaker model at dial 0.5; they separate nothing on accuracy, so what follows is about calls and dollars.

The playbook cuts calls per task. Learning against frozen: terminal 2.17 to 1.33, oolong 4.90 to 3.85, finance 3.45 to 2.40, legal 12.65 to 10.60. Only two packs adopted anything, though: oolong (16 of 20 candidates) and legal (5 of 21). On terminal and finance every candidate was rejected because its replay cost more than the baseline’s, so those learning runs are frozen runs with ratings on, and their call reduction is run-to-run variance, not learning. On oolong the second half of the learning run took 3.9 calls per task where the first half took 3.8: most of the gain arrived with the first adopted entries. On legal the halves are 13.7 and 7.5, the curve the design predicts.

The gate is the cost. Each candidate is replayed on three earlier tasks in both arms, six task runs per solved task, and those replays cost \$1.25, \$10.70, \$5.46 and \$16.63 on the four packs against \$0.04, \$1.92, \$0.63 and \$2.61 for the counted runs. At twenty tasks a pack the gate spends far more than the playbook saves. It earns its keep only over a long stream, and a cheaper gate (a smaller sample, a proxy model, a cached baseline arm) is the obvious next change.

Pack	Mode	Solved	Calls / task	Tokens / task	Dollars
terminal (6)	frozen	6/6	2.17	10 551	0.07
	learning	6/6	1.33	6 742	0.04
	plain	6/6	2.50	2 395	0.03
oolong-like (20)	frozen	20/20	4.90	30 778	1.51
	learning	20/20	3.85	41 991	1.92
	plain	20/20	1.00	3 084	0.39
finance-synthetic (20)	frozen	20/20	3.45	42 177	0.83
	learning	20/20	2.40	29 999	0.63
	plain	20/20	1.00	1 244	0.07
legal-synthetic (20)	frozen	20/20	12.65	62 452	1.69
	learning	20/20	10.60	58 315	2.61
	plain	20/20	1.00	1 475	0.11

Table 4: Claude Opus 5.5, 27 September 2026, one run per cell. Gate replays (not included): \$1.25 terminal, \$10.70 oolong, \$5.46 finance, \$16.63 legal.

The plain agent wins on cost here. Every context in these packs fits in one prompt (under 4 000 characters), so the baseline answers in one call with no tool use: \$0.39 against \$1.51 on oolong, \$0.07 against \$0.83 on finance, \$0.11 against \$1.69 on legal. The SQL abstraction pays for a catalog, a planner and a transcript of rendered results on every turn, and at this scale that is pure overhead. The claim of §1 is about work that does not fit in one call; these packs do not test it, and the numbers say so.

Per-clause calls are cheap and visible. On legal the model classified clause by clause with a prompt function on the `worker` alias, up to 26 calls a task at about \$0.10: the λ_f map-call doing what it is for, many small calls on a cheap model, priced before they run.

Costs drift up within a run. Per-task dollars on oolong learning rose from \$0.084 in the first half to \$0.107 in the second while calls stayed flat. Two things grow the prompt: the adopted playbook entries, and, as it turned out, the scratch tables every session created in the shared store, which the catalog listed to every later session (287 such tables after these runs). The second was a bug: it also changed the prompt fingerprint from task to task, so a recorded run could not be replayed from its own fixtures. Bench sessions now drop their tables when they finish.

7.3 Claude Haiku 4.5 on the harder packs

The harder packs ran on 1 October 2026 with every solver alias on `claude-haiku-4-5-20251001` and the `judge` alias on `claude-sonnet-5-5`. The sweep was stopped before it finished; Table 5 is what completed. Not in the table: `logbook-hard` in any mode (a frozen task took about 500 calls and over twenty minutes at Haiku’s pace, so the pack was capped at ten tasks and then stopped before the first row landed), and `coding` in plain mode (below). Gate replays cost \$1.43 on memo-rubric (11 candidates, 3 adopted) and \$1.04 on coding (2 candidates, 1 adopted); the counted runs cost \$3.36 in all.

These packs separate on accuracy. Where Opus 5.5 solved every task of the first four packs, Haiku solves 17% of the coding steps and 40 to 55% of the memos. The coding steps it solves are the first step of an episode; it solved no “extend” or “fix the bug report” step in either KLEENE mode.

Pack	Mode	Solved	Calls / task	Tokens / task	Dollars
coding (12)	frozen	2/12	4.58	48 375	0.87
	learning	2/12	2.67	29 910	0.47
memo-rubric (20)	frozen	8/20	19.95	49 043	1.30
	learning	11/20	3.80	23 041	0.37
	plain	8/20	3.15	10 764	0.35

Table 5: Claude Haiku 4.5 solvers with a Sonnet 5.5 judge, 1 October 2026, one run per cell; the sweep was stopped before `logbook-hard` and the coding plain run produced rows. The learning rows share the frozen rows’ memo (see text).

Learning beat frozen on memo-rubric, on both axes. 11/20 against 8/20, at 3.8 calls a task against 19.95 and \$0.37 against \$1.30. Two effects are mixed in. Three playbook entries were adopted, and the second half of the learning run solved 5 of 10 where the frozen run’s first half solved 3 of 10. But most of the frozen run’s calls are a failure mode the playbook happens to steer around: Haiku writes `llm_json` output schemas the API rejects (`additionalProperties` unset, `minItems` above 1, a bare `{"hours": "number"}`), and the statement retries until the call budget is gone; five frozen tasks took 22 to 141 calls each. The adapter now rewrites a model-written schema into the accepted subset before the request; these runs predate it.

Learning and frozen share the memo. The learning run came second in the same store, so a prompt identical to one the frozen run sent was answered from the memo for free. Two coding steps in the learning run finished with zero calls: they replayed the frozen run’s failed answers. The learning column’s calls and dollars are therefore not an independent measurement; the documentation now prescribes a fresh store per mode.

The plain baseline needs a protocol the model keeps. The agent asks for one JSON action per reply. Haiku answers with five to ten actions and invented tool output between them; the first extractor took the span from the first brace to the last, which never parsed, so on every coding task the model saw only a parse error and gave up after two turns (0/12 at \$0.45). The extractor now takes the first balanced object and runs it. With that, Haiku worked through the first coding task for all 30 turns without solving it (\$1.11) and the rerun was stopped before a row landed. On memo-rubric, where the task fits in one call, the plain agent matches frozen on accuracy at a quarter of the cost, and learning beats it by three tasks for the same money.

The judge wobbles. Sonnet 5.5’s first lines included “PASS (with caveat)”, “PASS (provisional)” and “PASS or FAIL?”, all counted as passes by the first parser. It now requires a first line that says PASS and does not mention FAIL. The memo frozen and learning runs predate that change; the plain run does not.

Two provider facts. Haiku rejects the `effort` parameter the adapter sent along with the rest of the model family, so under the default routing every `proxy` call failed until the adapter learned to omit it; and `claude-sonnet-5-5` was served on 1 October where it was not on 27 September.

7.4 Planner estimates against actuals

The Haiku run was the first made after `bench run` began tracing statements, so its store holds an estimate and an actual call count for each of the 1 028 statements the planner priced (Figure 4). Reading the plot: 670 statements made no calls and were estimated at none; 254 were estimated

at one call and made one; 82 were estimated at one and made none, which is the memo answering a repeated prompt. The planner never under-estimated by more than a call. Where it was wrong it was high, by up to two orders of magnitude on about a dozen statements, each a per-row call priced on a table’s row count that then failed early or was served from the memo. For budget refusal that is the safe direction; for the learned cost model it is the error to drive down.

7.5 What the numbers say

Taken together, the two runs support three statements and refuse a fourth. First, the planner, the budget and the trace work as designed on real models: estimates are conservative, refusals carry plans, per-clause calls land on the cheap alias, and every cost in this section was read from the store. Second, the playbook does reduce calls per task when something is adopted, and on the one pack that separates on accuracy it also raised solves, with the confounds stated. Third, the replay gate, at its current sample size, is the dominant cost of learning at pack scale. What the runs do not support is the headline cost-parity claim: wherever the context fits in one prompt the plain agent is cheaper, and the pack built to test the other case (`logbook-hard`, a 30 000-token context that has to be filtered and aggregated) has not yet produced a row.

8 Related work

Recursive language models. Zhang et al. [42] introduced the REPL-variable framing and the `llm_query` / `rlm_query` sub-calls that KLEENE maps onto `llm` and `rlm`; their evaluation suite (OOLONG, BrowseComp-Plus) is the headline comparison the `logbook-hard` pack is built toward. Wang [39] reproduced the result and documented the depth-two degradation that set KLEENE’s default depth cap. DSPy’s `dspy.RLM` [18] is the most used implementation; like the reference harness it is Python-based with fixed iteration and call caps rather than a priced plan. CodeAct [40] made the general case that executable code is a better action space than JSON tool calls; KLEENE agrees and narrows the language to one with an optimizer.

SQL and relational operators over language models. The closest systems embed model calls in a relational language and optimise them. LOTUS [29] defines semantic filter, map, join, aggregate and top-*k* operators with proxy cascades and statistical guarantees; KLEENE’s cascade rule has LOTUS’s shape with thresholds declared rather than calibrated. Palimpzest [23] and Abacus [31] run a Cascades-style [11] cost-based optimiser over model and prompt choices using sampled sentinel plans; KLEENE’s planner is simpler (eight rules, one cost model) but operates on the model’s own SQL at run time rather than on a pre-declared pipeline. ThalamusDB [17] adds `NLfilter` and `NLjoin` to DuckDB with progressive error bounds and call budgets. BlendSQL [10] is closest in spirit to CallSQL: a dialect with `LLMMap`, `LLMQA` and `LLMJoin` whose AST is rewritten so cheap SQL predicates run first; SUQL [24] exposes free-text primitives as lazily evaluated Postgres UDFs; Galois [32] treats the model itself as the database executing plan fragments; CAESURA [37] uses the model as the planner; Evaporate [2] synthesises extraction code instead of prompting per row, which is what a refined `CREATE FUNCTION ... AS SQL` amounts to. Liu et al. [25] reorder rows to share prompt prefixes in relational workloads, a rule KLEENE does not yet have. Scallop [22] makes foundation models foreign predicates in Datalog. Su et al. [36] give a taxonomy and benchmark of model-enhanced relational operators. What distinguishes KLEENE from all of these is that the *model writes the query*, turn by turn, with delegation, tools, budgets and learning in the same language, rather than a person writing a pipeline the model fills in.

Query optimisation. The planner’s machinery is classical: System R’s dynamic programming [33], bushy enumeration without cross products [28], semi-naïve fixpoint evaluation [3],

and the complexity results of Chandra and Merlin [6], Vardi [38], Aho and Ullman [1] and Dantsin et al. [8] that label each statement’s fragment. The name honours three of Kleene’s results [19–21]: the star as closure under repetition (what a recursive CTE computes), the fixed-point construction (how it computes it), and the recursion theorem (a session opening a child session is the same loop referring to itself).

Continual harnesses and curricula. The loop’s learnability target follows Absolute Zero Reasoner [43]; its generators follow Reasoning Gym [35] and the 3-SAT phase transition [27]; its ratings are Bradley–Terry [5]. The replay gate is the harness-level analogue of the default-FAIL test contract common to long-running agent harnesses: nothing is adopted that has not beaten the current version on held tasks.

9 Limitations and future work

- **One run per cell.** Every number in §7 is a single run. On twenty tasks accuracy moves in steps of five points, and a difference of one task is not a result. Repeated runs with fresh stores are the first thing to add.
- **Saturated and synthetic packs.** The first four packs separate nothing on accuracy under Opus 5.5, and the synthetic packs are generated, so a solver that learns the generator’s shape will look better than one meeting the real datasets. Real OOLONG, Terminal-Bench, SWE-bench and Harvey LAB imports are the remedy; `bench import-lab` exists and an OOLONG importer is in review.
- **The long-context case is untested.** `logbook-hard`, the pack built to test the claim of §1, has not produced a row. It needs a stronger solver than Haiku or a cap of ten tasks per mode.
- **Coding is weak.** Two of twelve steps on Haiku in both KLEENE modes, and the plain baseline none. The failing rows show the model declaring `FINAL` after two or three turns while hidden tests still fail; a finish gate that refuses `FINAL` while a code task’s tests fail, and a native edit-run-read tool loop with `CallSQL` as one tool among others, are the changes this points at.
- **Shared memo across modes.** The Haiku learning column is not an independent measurement (§7.3). Fresh store per mode, or the memo excluded from the gate’s accounting, fixes it.
- **The gate’s cost.** Six task runs per solved task dominates a learning run at pack scale. A smaller sample, a proxy model for the replay, or a cached baseline arm are the candidates.
- **Declared thresholds, per-session cost model.** Cascade thresholds are declared, not calibrated from a sample as in LOTUS; sampled selectivities persist only within a session rather than as versioned tables.
- **The plain baseline.** It parses JSON actions from text rather than using native tool calling, and a model that writes several actions per reply gets only its first one run. A native tool-calling baseline would make the cost-parity comparison cleaner.
- **Schema rewriting is partial.** Model-written `llm_json` schemas are rewritten into the provider’s accepted subset only for the shapes seen so far.

Beyond these, the design names measurements the harness supports but no run has yet exercised: transfer (learn on a prefix with `-limit`, run the rest frozen on the same store, compare against a never-learned store), reverts through `learn playbook`, and the plan-space plot over the shapes models actually write.

10 Reproducibility

The repository is a Rust workspace; DuckDB builds from source once (about ten minutes). Every claim in Table 3 is a test:

```
cargo test --workspace --all-features
cargo run -- repl < demos/planner/three_way.sql      # the plan space, live
cargo run -- learn run --tasks 20 --generators puzzle,corpus
cargo run -- bench run tasks/terminal --mode frozen --record fixtures/terminal
cargo run -- bench report
cargo run -- bench plot plots/
```

The per-task rows of §7.2 are `plots/evals.csv` and those of §7.3 are `plots/haiku-2026-10-01/evals.csv`, with the `bench report` output, the router file and the SVGs beside them. The model replies of the Opus run were recorded but are not checked in, because the catalog bug of §7.2 means they cannot replay the run; later runs record with `-record` and replay with `-replay`. Prices are the router’s list rates at the time of each run: \$4 / \$20 per million input / output tokens for Opus 5.5, \$2 / \$10 for Sonnet 5.5.

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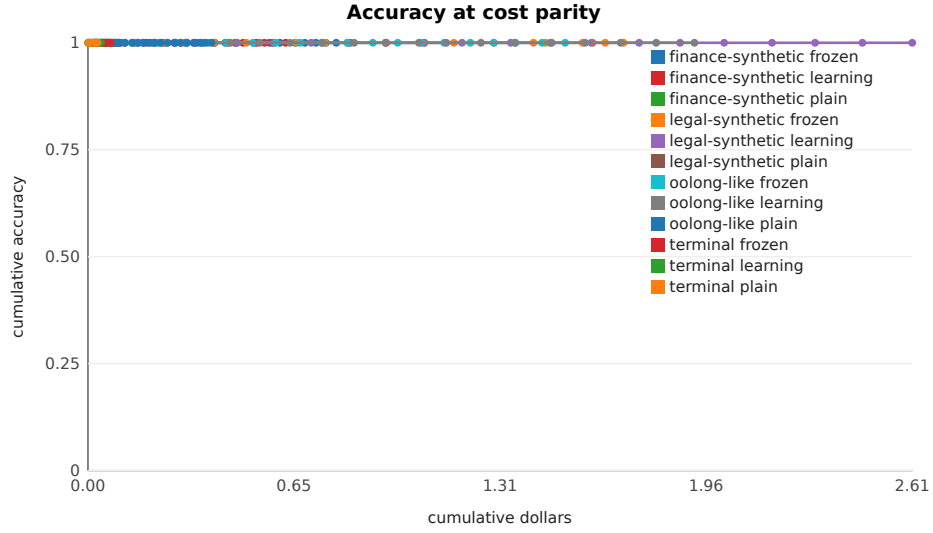
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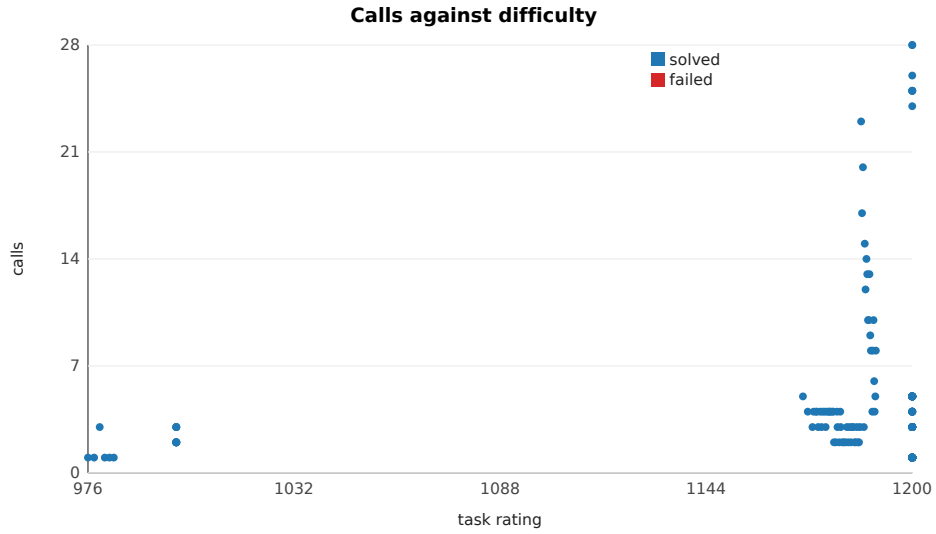
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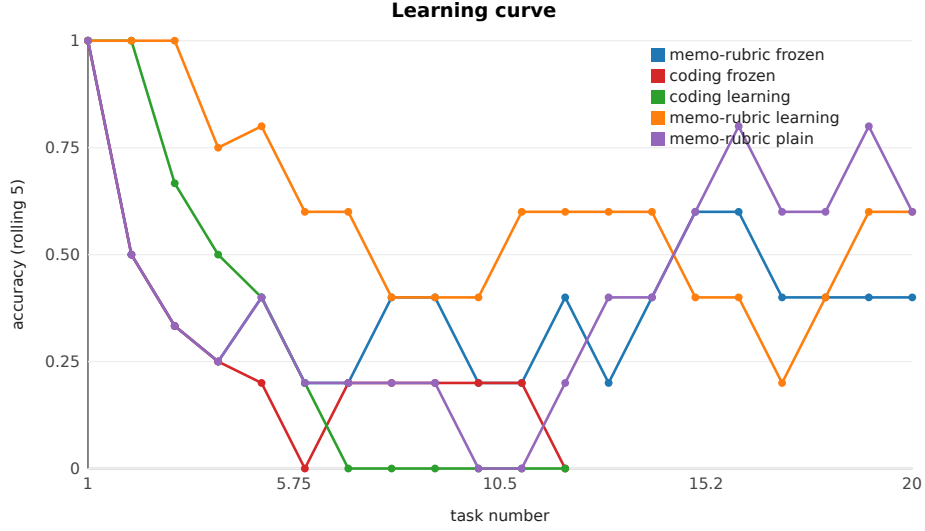


(a) Cumulative accuracy against cumulative dollars.

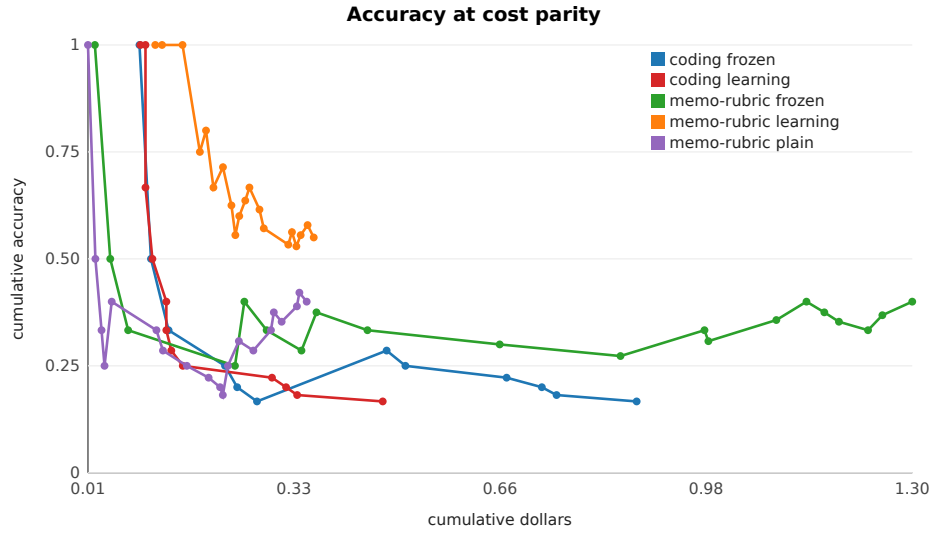


(b) Calls per task against the task's rating.

Figure 2: The Opus 5.5 run. Every line reaches accuracy 1; the plot separates the modes on cost only, with the plain agent leftmost on every pack.



(a) Rolling-5 accuracy by position in the stream.



(b) Cumulative accuracy against cumulative dollars.

Figure 3: The Haiku 4.5 run on `memo-rubric` and `coding`. On `memo-rubric` the learning line ends above frozen and plain at a fraction of frozen’s spend; on `coding` both KLEENE lines fall to zero after the first step of each episode.

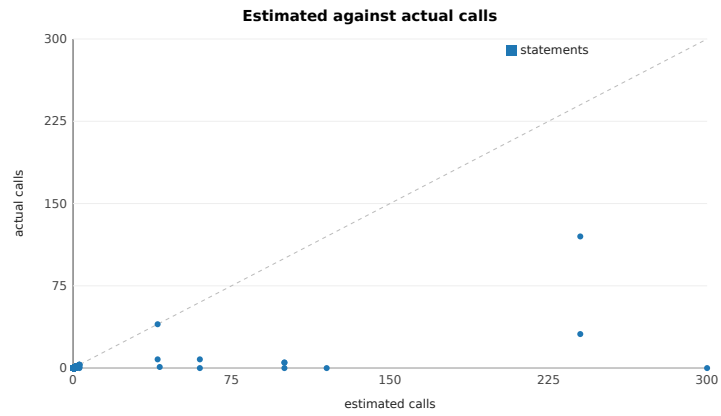


Figure 4: Estimated against actual calls per statement, Haiku run. Points on the diagonal are exact; the outliers lie below it (over-estimates).